

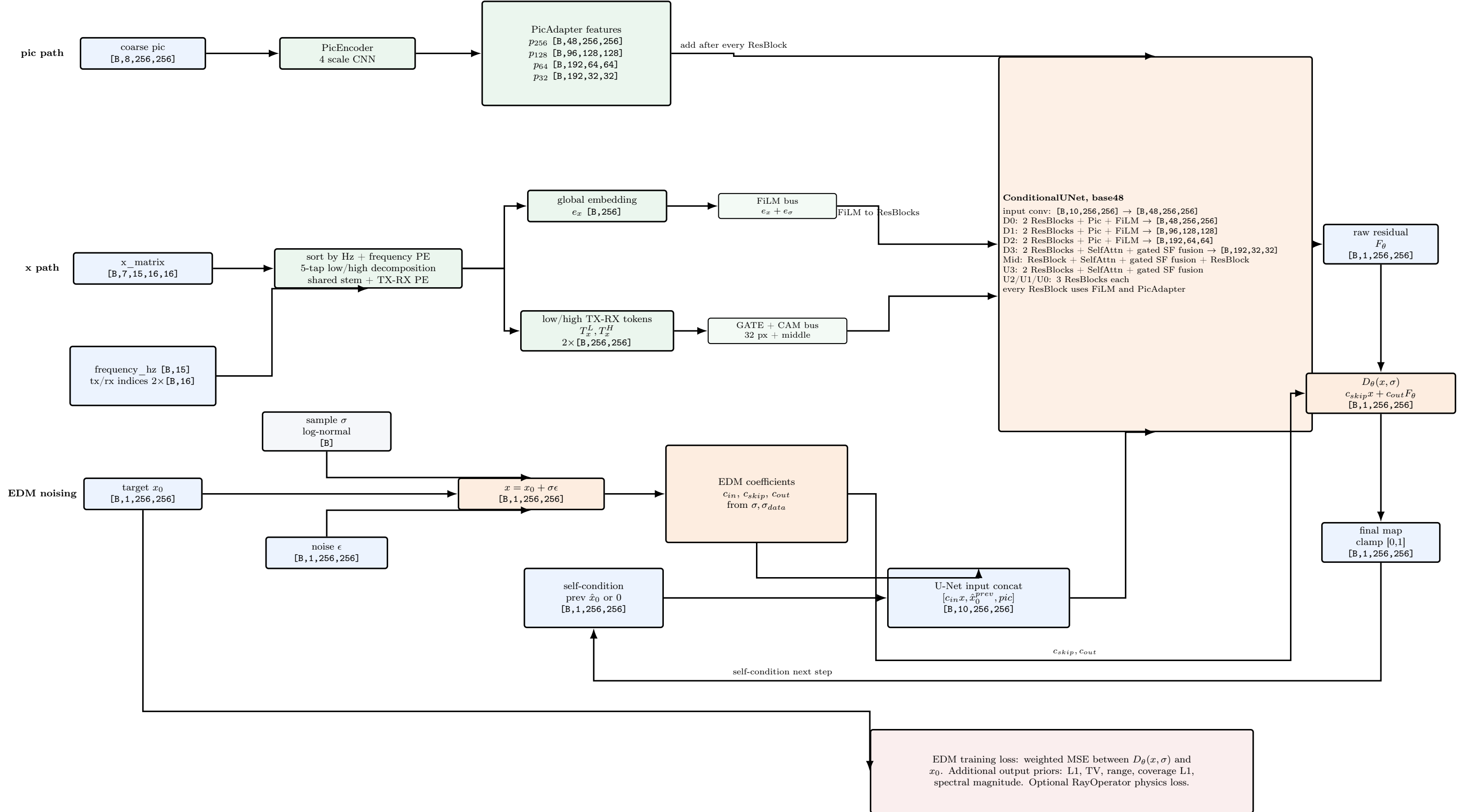
EDM Conditional Diffusion Network for Ultrasonic Defect Map Inversion

Follows `dataset_a_256_base48_edm.yaml1`: image size 256, pic channels 8, x channels 7, frequencies 15, base channels 48.

All arrows are routed as horizontal/vertical orthogonal polylines. Tensor order: images $[B, C, H, W]$; x-matrix $[B, C, F, TX, RX]$; x tokens $[B, N, D]$.

Training view. EDM uses continuous noise scale σ . The target x_0 is corrupted as $x = x_0 + \sigma\epsilon$. The network predicts a denoised map through EDM preconditioning, not a DDPM v or ϵ target.

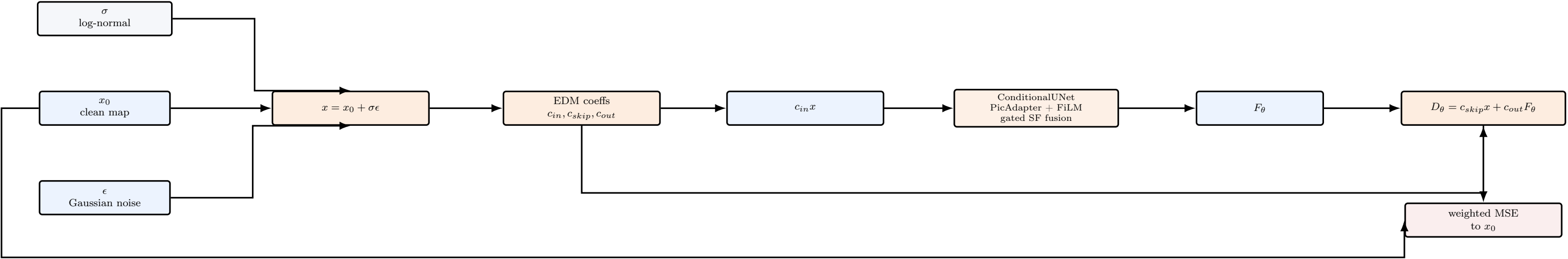
Sampling view. Karras sigmas are traversed from σ_{max} to 0. Each step computes $\hat{x}_0 = D_\theta(x, \sigma)$ and applies Euler/Heun update. Repeat the trajectory for K independent noises; the posterior mean is the default prediction (page 6).



EDM Preconditioning and Tensor Shapes

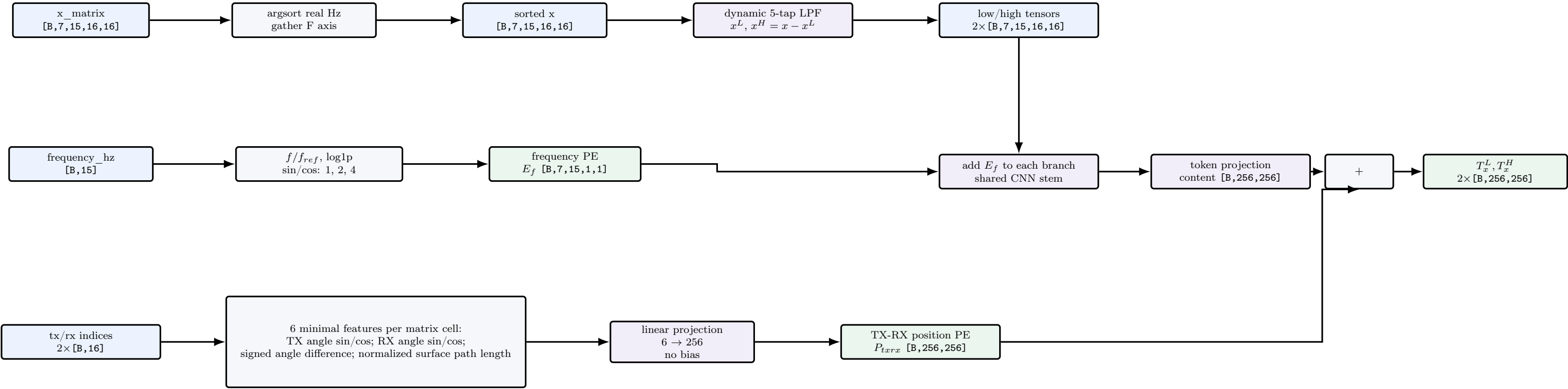
EDM preconditioning is combined with dynamic physical-frequency decomposition and gated spatial-frequency fusion.

Stage	EDM formula / operation	Shape	Purpose
Clean target	x_0	[B, 1, 256, 256]	Ground-truth defect-depth label.
Noise scale sampling	$\log \sigma \sim \mathcal{N}(p_{mean}, p_{std}^2)$, clamp to $[\sigma_{min}, \sigma_{max}]$	[B]	Continuous EDM training noise level. Base48 config uses $\sigma_{data} = 0.5$, $\sigma_{min} = 0.002$, $\sigma_{max} = 80$, $p_{mean} = -1.2$, $p_{std} = 1.2$.
Noisy image	$x = x_0 + \sigma \epsilon$	[B, 1, 256, 256]	No forward Markov chain or beta schedule is used.
Coefficients	$c_{in} = 1/\sqrt{\sigma^2 + \sigma_{data}^2}$; $c_{skip} = \sigma_{data}^2/(\sigma^2 + \sigma_{data}^2)$; $c_{out} = \sigma \sigma_{data}/\sqrt{\sigma^2 + \sigma_{data}^2}$	broadcast to image dims	Stabilizes network input/output scale across noise levels.
Network input	$\text{concat}(c_{in}x, \hat{x}_0^{prev}, pic)$	[B, 10, 256, 256]	Same channel count as DDPM version, but the first channel is preconditioned noisy input.
Noise embedding	$e_\sigma = \text{TimeEmbedding}(\log \sigma/4)$	[B, 256]	Reuses the U-Net time-embedding module with EDM's continuous log-sigma coordinate.
Condition vector	$c = e_x + e_\sigma$	[B, 256]	FiLM condition for every ResBlock.
Physical-frequency tokens	sort by real Hz; inject frequency PE before aggregation; dynamic 5-tap LPF; shared stem; add TX-RX physical PE	T_x^L, T_x^H : $2 \times [\text{B}, 256, 256]$	Each token represents one TX-RX cell; all 15 real frequencies are encoded before tokenization.
Raw U-Net output	$F_\theta(c_{in}x, \hat{x}_0^{prev}, pic, e_x, T_x^L, T_x^H, \sigma)$	[B, 1, 256, 256]	Network residual after gated spatial-frequency fusion.
Denoised output	$D_\theta(x, \sigma) = c_{skip}x + c_{out}F_\theta$	[B, 1, 256, 256]	Direct clean-map prediction.
Training loss	$E_{x_0, \epsilon, \sigma} [w(\sigma) \ D_\theta(x, \sigma) - x_0\ _2^2]$; $w = (\sigma^2 + \sigma_{data}^2)/(\sigma \sigma_{data})^2$	scalar	Replaces DDPM's MSE on v or ϵ .



Physical Frequency and TX-RX Position Encoding

Real Hz is encoded before frequency aggregation; cylindrical TX-RX geometry is added after token projection.

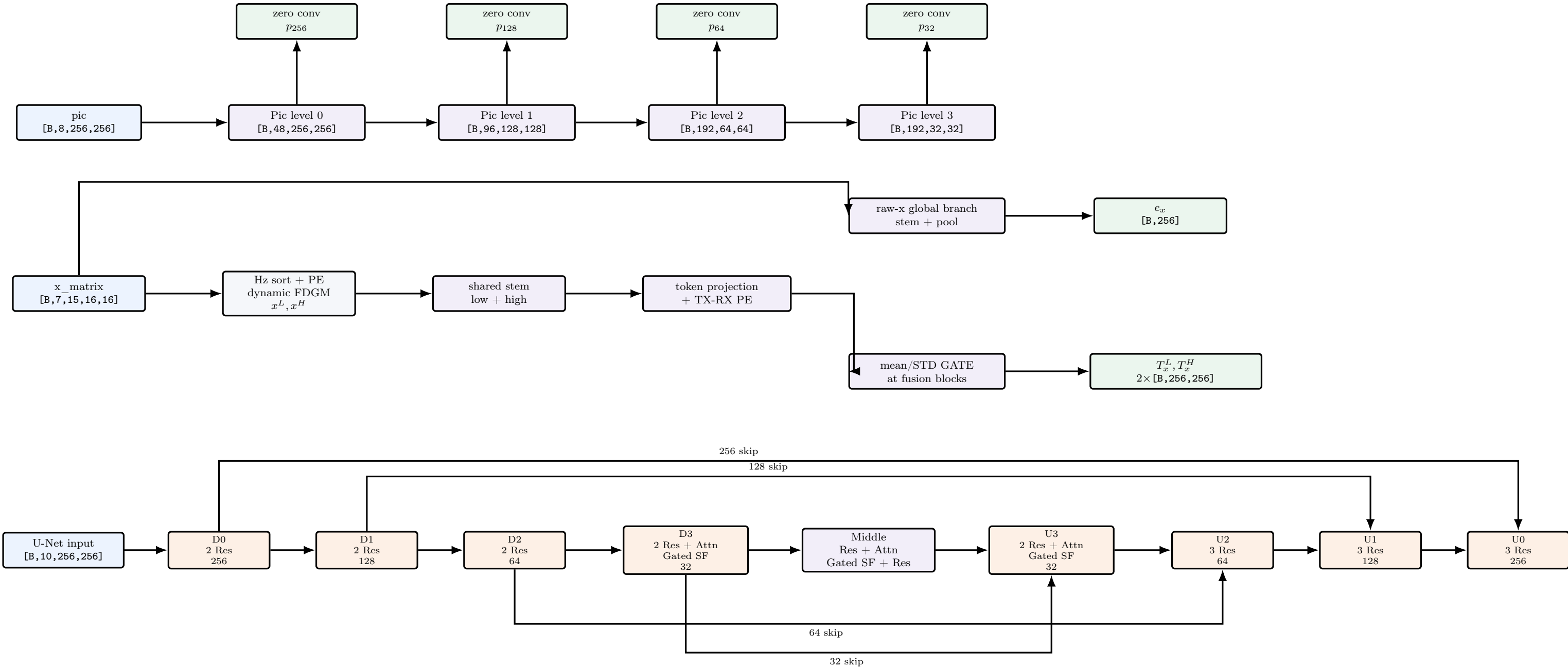


Token meaning. Token $n = tx_row \times 16 + rx_column$ is one TX-RX matrix cell. It summarizes all 15 frequency slices and therefore is not a single-frequency token.

Domain meaning. The same P_{txrx} is added to low and high content. Separate learned domain embeddings are added later inside each fusion block.

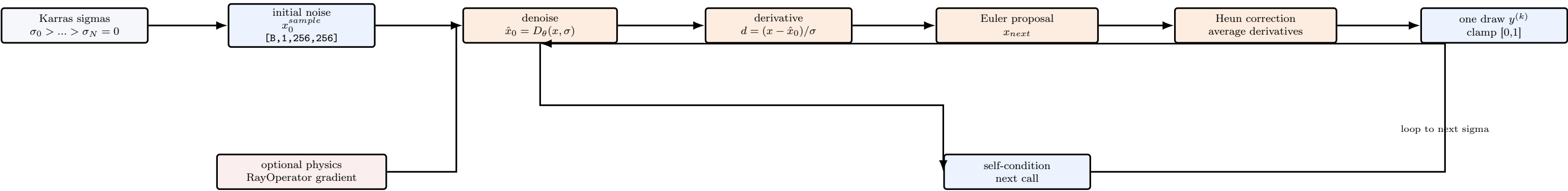
Shared Conditional Backbone

PicEncoder/PicAdapter and U-Net scales remain; the x path produces physically located low/high tokens.



EDM Single-Trajectory Sampling and Physics Guidance

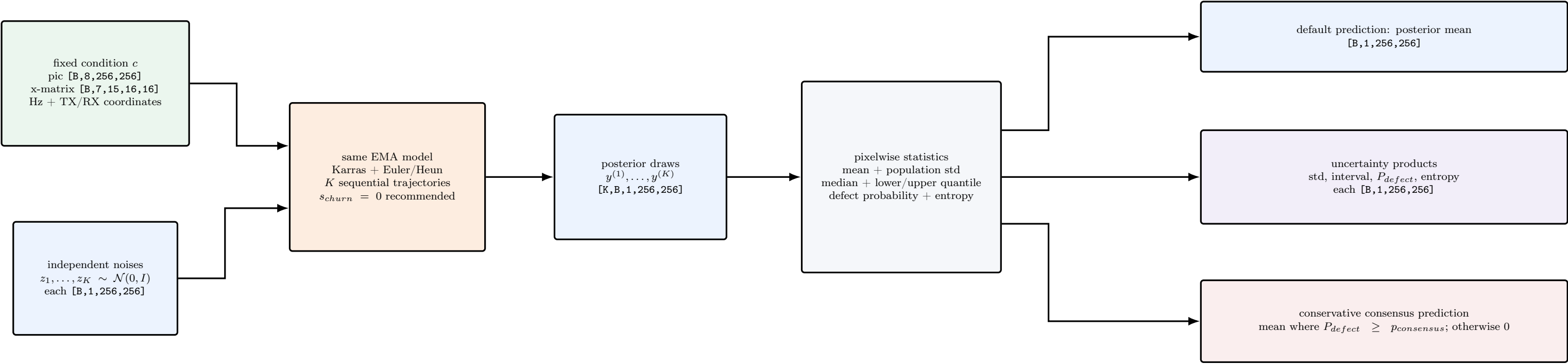
Component	EDM behavior in code
Sigma schedule	<code>karras_sigmas(steps)</code> creates a monotonically decreasing sequence from σ_{max} to σ_{min} and appends 0. The base48 config uses 32 sampling steps by default.
Initial image	Posterior draw k starts from independent $z_k \sim \mathcal{N}(0, I)$ and $x_k = \sigma_{max} z_k$ with shape <code>[B,1,256,256]</code> .
First denoise	Compute $\hat{x}_0 = D_\theta(x, \sigma)$ using the same pic and <code>x_matrix</code> conditions as training.
ODE derivative	$d = (x - \hat{x}_0)/\sigma$. This replaces DDPM posterior mean/variance formulas.
Euler/Heun update	First predicts $x_{next} = x + (\sigma_{next} - \sigma)d$, then optionally recomputes denoising at σ_{next} and averages derivatives.
Self-conditioning	The current denoised output is fed as $\hat{x}_0^{prev}$ in the next denoising call.
Physics guidance	After a configured fraction of sampling steps, $\hat{x}_0$ can be adjusted by the gradient of frequency-aware RayOperator consistency loss before forming the ODE derivative.



Difference from diffusion	Summary
Training target	Diffusion predicts v by default and recovers x_0 from y_t, t, v . EDM directly trains the denoised output $D_\theta(x, \sigma)$ against x_0 .
Noise variable	Diffusion uses integer timestep t and beta schedule. EDM uses continuous σ and log-normal training noise distribution.
Sampler	Diffusion uses DDPM/DDIM formulas. EDM uses Karras sigma schedule and ODE-style Euler/Heun updates.
Backbone	EDM now adds dynamic physical-frequency decomposition and gated SF fusion; PicEncoder/PicAdapter, FiLM, U-Net scales, and RayOperator remain.

Single-Model Multi-Sample Posterior and Uncertainty Maps

The condition and network weights remain fixed; only the initial EDM noise changes across posterior draws.



Fixed evidence path. Every trajectory uses the complete page-1 pipeline: physical-frequency/TX-RX encoding, low/high token fusion, PicAdapter, FiLM, and the same EDM denoiser weights.

Sparse hallucination rule. A bright pixel present in only a small fraction of draws has low P_{defect} and is removed from the consensus map. The posterior mean remains the default continuous prediction. A systematic false positive shared by all draws cannot be detected by this rule.

Output / metric	Definition and interpretation
Posterior mean	$\mu = (1/K) \sum_k y^{(k)}$. Saved as *_prediction.npy ; this is the default result for MAE, RMSE, SSIM, and volume error.
Uncertainty map	$s = \sqrt{(1/K) \sum_k (y^{(k)} - \mu)^2}$. It measures disagreement between conditional samples, not inverse path coverage and not complete model-parameter uncertainty.
Defect probability	$P_{defect} = (1/K) \sum_k \mathbf{1}[y^{(k)} \geq \tau_{defect}]$. Binary entropy is highest when samples disagree about whether a defect exists.
Calibration checks	Report empirical interval coverage, interval width, CRPS when raw draws are retained, uncertainty-error Spearman correlation, Brier score, and uncertainty versus path coverage/reliability.
Recommended K	Use $K = 16$ for routine inspection and $K = 32$ for final stability checks. $K = 2$ or 4 is only a smoke-test setting.